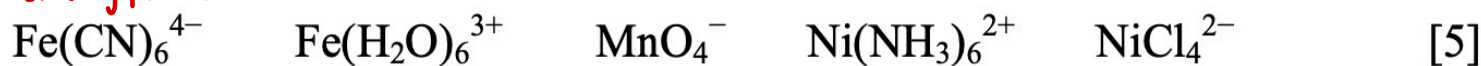


Predict the number of unpaired electrons in the following species, explaining your reasoning in terms of simple energy level diagrams.

strong field



Td always high spin
 $\Delta_T < \Delta_O$

0 upe⁻

5 upe⁻

0 upe⁻

2 upe⁻

2 upe⁻

Redox inorganic chemistry



e.g. Write a balanced equation for the oxidation of Fe^{2+} by MnO_4^- in acid solution.



e.g. Write a balanced equation for oxidation of Zn metal by MnO_4^- in acid solution.



Spontaneity of redox reactions

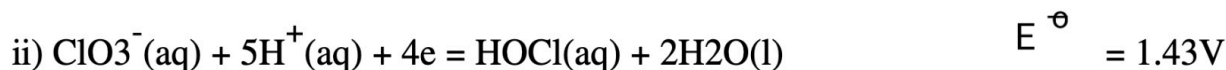
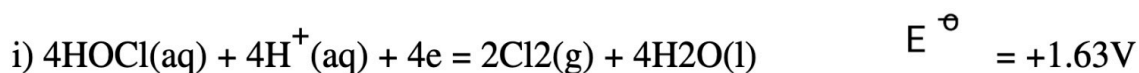
Thermodynamics $\longrightarrow$ "spontaneous" $\begin{cases} \text{thermodynamically favourable} \\ \text{feasible} \end{cases}$

For a spontaneous reaction $\Delta G < 0$ ($K > 1$)

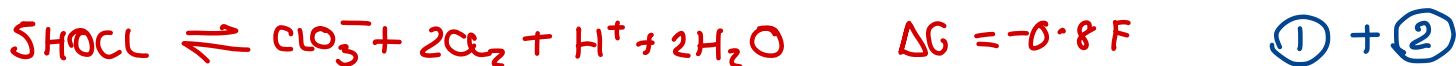
Note that $\Delta G = -RT \ln K$

For a half-reaction $\Delta G = -nFE^\ominus$ — standard electrode potential
number of e^- transferred Faraday constant

(d) Given the half equations:



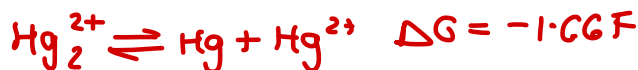
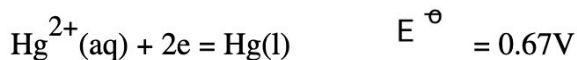
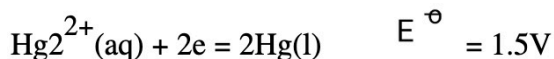
calculate whether or not hypochlorous acid **disproportionates** under acid conditions



Thermodynamically unstable WRT disproportionation but kinetically stable

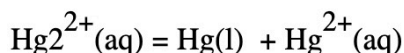
$\Delta G < 0 \therefore$ disproportionates (but kinetics prevents it)

2. Standard Electrode potentials for mercury are as follows:



$\Delta G < 0 \therefore$ disproportionates

Calculate the equilibrium constant for the disproportionation



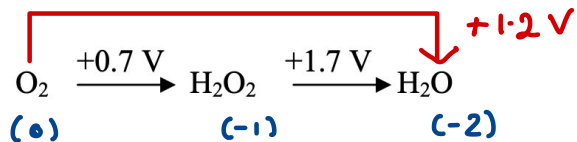
$$\Delta G = -RT \ln K \therefore K = \exp\left(-\frac{\Delta G}{RT}\right)$$

$$K = \exp\left(-\frac{-1.66 \times 96500}{8.314 \times 298}\right) \approx 1.2 \times 10^{28}$$

$K > 1 \therefore$ feasible

Answer all parts.

The Latimer diagram for oxygen in acidic solution is given below:



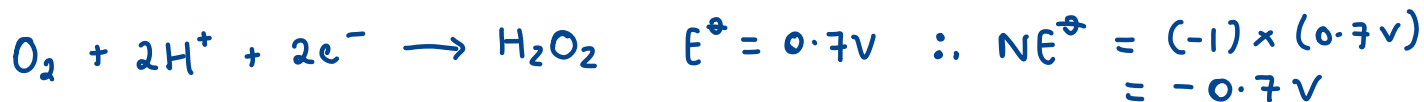
Oxidation numbers decrease from left to right and E° written above the arrow

(i) Sketch a Frost (oxidation state) diagram for oxygen in acidic solution. [4]

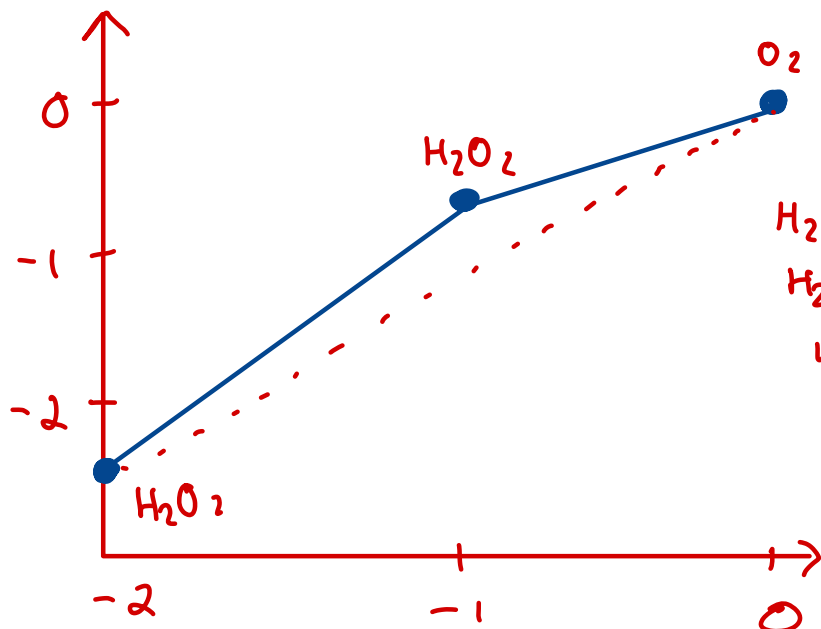
(ii) Using your diagram, demonstrate that H_2O_2 is unstable with respect to disproportionation in acidic solution and provide a balanced chemical equation describing the disproportionation reaction. Calculate the equilibrium constant for this process at 298K. [6]

$NE^\circ (X(N)/X(O))$
oxidation number

$$[RT/F = 0.026 \text{ V}; (RT/F) \ln 10 = 0.059 \text{ V at 298 K}]$$



$$\Delta G = -nFE^\circ \quad \therefore -4.8F = -4F E^\circ \quad E^\circ = 1.2 \text{ V} \quad NE^\circ = -2 \times 1.2 \text{ V} = -2.4 \text{ V}$$



H_2O_2 lies above the line joining H_2O and O_2 $\therefore \text{H}_2\text{O}_2$ unstable with respect to disproportionation